

BACKEND CURRICULUM (BEGINNER LEVEL)

Learn Backend by Building (Node + Express + MongoDB)

1. **Duration:** 3 Weeks
2. **Schedule:** 3 days/week × 2 hours/day
3. **Total:** 18 hours
4. **Teaching Style:** Step-by-step, practical, zero assumptions

FINAL OUTCOME (Very Important)

By the end of this program, students will be able to say:

"I built the backend of a real event management system using Node.js, Express, TypeScript, and MongoDB — handling users, events, registrations, payments, and attendance."

TECH STACK (BEGINNER-FRIENDLY)

1. **Runtime:** Node.js
2. **Language:** TypeScript
3. **Framework:** Express.js
4. **Database:** MongoDB (Mongoose)
5. **Tools:** Postman, dotenv
6. **Auth:** JWT (basic level)

WEEK 1 — BACKEND BASICS (NO RUSH)

Day 1 (2 Hours): What Backend Is & Setting the Ground

Goals

Remove fear of backend.

Explain backend in plain language.

Topics (Simple Language)

What is backend?

What happens when you submit a form?

Client vs Server.

What is an API?

What is Node.js?

What is Express?

Hands-On

Install Node.js.

Create first Express server.

Result: **GET** / → “Backend Running”.

Assignment

Explain backend using a WhatsApp analogy.

Day 2 (2 Hours): TypeScript for Absolute Beginners

Goals

Use TypeScript without confusion.

Topics

Why TypeScript?

Types: **string**, **number**, **boolean**.

Interfaces (simple).

Type safety in APIs.

Hands-On

Convert Express app to TypeScript.

Create typed request & response.

Assignment

Create a typed object for an **Event**.

Day 3 (2 Hours): MongoDB Fundamentals (No SQL Confusion)

Goals

Understand databases visually.

Topics

What is a database?

MongoDB vs SQL (simple comparison).

Collections & documents.

Mongoose basics.

Hands-On

Connect MongoDB.

Create first schema: **User**.

Save & fetch data.

Assignment

Draw database on paper.

WEEK 2 — BUILDING CORE FEATURES

Day 4 (2 Hours): User Authentication (Very Basic)

Goals

Understand login systems.

Topics

What is authentication?

Password hashing.

JWT (high level only).

Hands-On

Register organizer.

Login organizer.

Protect a route.

Assignment

Explain JWT in simple words.

Day 5 (2 Hours): Event Management (Main Feature)

Goals

Create real product functionality.

Topics

RESTful routes.

Controllers & services.

Hands-On

Create Event.

Get Events.

Organizer owns events.

Assignment

Add validation rules.

Day 6 (2 Hours): Public Event Registration (Critical)

Goals

Handle public users safely.

Topics

Public vs private routes.

Input validation.

Prevent duplicate entries.

Hands-On

Register attendee.

Phone number required.

Save unpaid status.

Assignment

Improve error messages.



WEEK 3 — REAL-WORLD LOGIC & POLISH



Day 7 (2 Hours): Payments Logic (MVP Reality)

Goals

Understand real payment workflows.

Topics

Payment states.

Why manual confirmation exists.

Hands-On

Confirm payment manually.

Payment records per event.

Assignment

Draw payment flow.



Day 8 (2 Hours): Attendance & Check-In Logic

Goals

Solve event-day problems.

Topics

Attendance tracking.

Prevent duplicate check-ins.

Hands-On

Check-in attendee.

Search by phone number.

Assignment

Explain why phone is better than email for check-in.



Day 9 (2 Hours): Backend Review, Cleanup & Demo

Goals

Build confidence.

See the big picture.

Topics

Folder structure.

Error handling.

Environment variables.

Deployment overview.

Hands-On

Review full Eventzity backend.

Postman demo of all APIs.

Final Project

Each student explains one backend feature they built.